

Shyam Charan Reddy Karra

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SKILLS

- **Languages:** C++, Java, Python, Kotlin, HTML, CSS, JS
- **Frameworks:** Bootstrap, Tailwind CSS, React, Vue
- **Tools/Platforms:** Leet Code, Geeks for geeks, GitHub, Excel, VS Code
- **Skills:** Problem-Solving, Logical Reasoning, Adaptability

PROJECTS

Interactive Meme Editor & Generator

Apr'25 - Jun' 25

- Designed and developed a responsive Single Page Application (SPA) to facilitate seamless meme creation, managing cross-component state and data persistence via dynamic routing and URL parameters.
- Implemented interactive canvas manipulation using react-draggable for coordinate-based text positioning and integrated html2canvas to rasterize DOM elements into downloadable high-quality image files.
- Integrated the Imgflip REST API to fetch and randomize 100+ global templates asynchronously, while engineering a custom upload feature using File Blobs for instant client-side previews. Tech: ReactJS, Tailwind CSS, Firebase, Imgflip API.

Full-Stack AI Sentiment & Business Intelligence Platform

Sep'25 - Nov' 25

- Engineered a high-performance FastAPI backend integrating a dual-engine machine learning pipeline (HuggingFace RoBERTa and VADER) with an O(1) heuristic boundary for low-latency, real-time sentiment classification.
- Developed a custom Explainable AI (XAI) Token-Valence Proxy to bypass SHAP computational limits for synchronous keyword attribution, paired with dynamic LDA topic modeling to isolate recurring product pain points.
- Designed a responsive React & Tailwind frontend dashboard incorporating an LLM-powered agentic layer (Gemini API) to automatically synthesize scraped review datasets, sentiment profiles, and topics into actionable executive summaries.

BharatSync Tourism & Heritage App

Mar'26 - May' 26

- Designed and developed an offline-first Android application promoting local tourism using Kotlin and Jetpack Compose, managing robust data persistence and local state through Room Database and Hilt dependency injection.
- Implemented an interactive mapping and scanning module by integrating OpenStreetMap (osmdroid) for location-based heritage tracking, and utilized Google ML Kit with CameraX for real-time computer vision identification of local handicraft tags.
- Integrated the Groq REST API via Retrofit to asynchronously fetch and generate customized, AI-driven travel itineraries using Llama-3.3, highlighting local artisans and GI-tagged products. Tech: Kotlin, Jetpack Compose, Room, Retrofit, ML Kit, Groq API.

Spotify Unsupervised Learning & Cohort Discovery

Apr'26 - May' 26

- Designed and developed a multi-algorithm unsupervised machine learning pipeline to analyze complex Spotify audio features, managing dataset normalization and discovering distinct musical cohorts within a 1,600+ track discography.
- Implemented diverse clustering architectures including K-Means, Agglomerative, DBSCAN, and Gaussian Mixture Models (GMM), integrating Silhouette analyses, k-distance graphs, and BIC/AIC evaluation for mathematically rigorous hyperparameter optimization.
- Engineered high-dimensional data visualizations using PCA and t-SNE embeddings to map topological cluster boundaries, while validating unsupervised cohort separation via supervised Random Forest feature importance modeling. Tech: Python, Scikit-learn, Pandas, Seaborn.

CERTIFICATES/CERTIFICATIONS

- Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate Mar'26 - Apr'26
- Summer Training in Full Stack — Lovely Professional University Apr'25 - Jun' 25
- Build Generative AI Apps and Solutions with No-Code Tools —Infosys SpringBoard Aug'25 - Sep' 25
- ChatGPT-4 Prompt Engineering: ChatGPT, Generative AI & LLM — Infosys SpringBoard Aug'25 - Sep' 25

TRAINING

- Completed an intensive Full Stack Development course in React and Node organized by the Centre for Professional Enhancement at Lovely Professional University.
- Mastered building scalable web applications by leveraging React for frontend interfaces and Node.js for server-side logic.
- Successfully secured an 'A' grade for high-level performance , culminating in the development of a functional full-stack project that demonstrated technical proficiency.

ACHIEVEMENTS

- Secured a 5-Star Badge in C++ on Hacker Rank by solving diverse algorithmic challenges across varying difficulties. Demonstrated mastery of Data Structures and STL by engineering solutions optimized for time and space complexity.
- Advanced through two competitive judging rounds during the **InnovateX** 24-hour hackathon, successfully building a functional prototype while demonstrating high-pressure problem-solving and technical execution within a strict development sprint.

EDUCATION

- [Lovely Professional University](#) Punjab, India
Bachelor of Technology - Computer Science and Engineering; *Since October 2023*
- [Sri Chaitanya Junior College](#) Telangana, India
Intermediate; Percentage: 96.4% *April 2020 - March 2022*
- [Sri Chaitanya High School](#) Telangana, India
Matriculation; CGPA: 10 *April 2019 - March 2020*